

AIR POLLUTION CONTROL DIVISION
Field Inspection Report

COUNTY NUMBER: 013 SOURCE NUMBER: 0003 DATE: 6-9-00
 COMPANY: Southdown, Inc. INSPECTOR: Thomas Lovell
 SITE LOCATION: 5134 Ute Highway, Lyons COUNTY: Boulder
 CONTACT PERSON: Steve Mossberg TIME:
 TELEPHONE NUMBER: (303) 823-2113 AIRS UPDATE:
 INSPECTION: ROUTINE[] COMPLAINT[] FA[] STACK TEST [X] PERMIT NUMBER: 95OPBO082
 HOURS: PROTOCOL: 3.0 INSPECTION: 8.0 TRAVEL: 4.0 REPORT: 8.0 TOTAL: 23.0
 EMISSION POINTS SUBJECT TO NSPS: Cement Plant (Subpart F)

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On April 26 and June 9, 2000, stack testing was performed at the Southdowns Cement plant in Lyons. The source tested is the Raw Materials Dryer located at the site (AIRS point 002). The source was tested to demonstrate compliance with its permitted emission limits for particulate, lead, NO_x, CO, SO₂, and VOC. The testing in April showed this source to be exceeding its emission limit for particulate. The particulate and lead tests were repeated on June 9. The results of the testing are summarized in the table below.

	Run 1	Run 2	Run 3	Average
Kiln Feed Rate (tons/hr)	150.28	150.28	150.02	150.19
Stack Flow (dscfm)	33000	30900	31100	31700
Oxygen (%)	2.0	2.0	2.0	2.0
Particulate Emissions (gr/dscf)	0.008	0.012	0.007	0.009
Particulate Emissions (lbs/hr)	2.1	3.1	1.8	2.3
Particulate Emissions (tons/yr)	7.5	11.0	6.1	8.2
Lead Emissions (lbs/hr)	0.00055	0.00030	0.00055	0.00047
Lead Emissions (tons/yr)	0.0019	0.0011	0.0019	0.0016
NO _x Emissions (ppm)	2.4	2.7	2.9	2.7
NO _x Emissions (lbs/hr)	0.56	0.60	0.64	0.60
NO _x Emissions (tons/yr)	2.0	2.1	2.2	2.1
CO Emissions (ppm)	6.3	11	11	9.4
CO Emissions (lbs/hr)	0.90	1.44	1.45	1.26
CO Emissions (tons/yr)	3.2	5.0	5.1	4.4
SO ₂ Emissions (ppm)	0.2	0.8	1.1	0.7
SO ₂ Emissions (lbs/hr)	0.06	0.24	0.34	0.21
SO ₂ Emissions (tons/yr)	0.21	0.84	1.2	0.74
VOC Emissions (ppm wet)	2.8	2.7	2.9	2.8
VOC Emissions (lbs/hr)	0.69	0.64	0.70	0.67
VOC Emissions (tons/yr)	2.4	2.2	2.4	2.3

The results of the second set of tests for particulate and lead are given in the above table above; all other numbers are from the first test. Emissions of lead during the first test were similar to the first set (0.0046 lbs/hr in April, 0.00467 lbs/hr in June). Emissions of particulate during the first test were 15.3 lbs/hr or 53.6 tons per year. Tons per year values in the table above are based on 7000 hours per year of operation, the operational hour limit on the permit.

Action #035

The results of the testing show the dryer to be in compliance with its permitted emission limits of 22.8 tons per year of particulate (there is also a PM₁₀ limit of 6.5 lbs/hr and 22.8 tons per year), 1.6 tons per year of lead, 13.9 tons/yr for NO_x, 57.3 tons/yr for CO, 36.7 tons/yr for SO₂, and 144.8 tons/yr for VOC. Natural gas was burned during the testing, since coal was not used as fuel during the year.

A compliance order was signed in October for this point. The order requires the source to perform another particulate test within one year. If the results are more than 50% of the permit limit, the source will have to perform another particulate test within 3 years of the effective date of the order.

On February 15, 2000, stack testing was performed on the cement kiln at the site. The kiln stack was tested to determine compliance with the permit and NSPS limits for particulate. The results of the testing are summarized in the table below.

	Run 1	Run 2	Run 3	Average
Kiln Feed Rate (tons/hr)	112.7	109.4	104.6	108.9
Stack Flow (dscfm)	134000	139000	135000	136000
Oxygen (%)	10.8	10.0	10.5	10.4
Particulate Emissions (gr/dscf)	0.0032	0.0012	0.0021	0.0022
Particulate Emissions (lbs/hr)	3.7	1.4	2.5	2.5
Particulate Emissions (tons/yr)	15.1	5.6	10.0	10.2
Particulate Emissions (lbs/ton)	0.033	0.013	0.024	0.023

Tons per year values in the table above are calculated based on the permit limit of 8064 hours per year of operation. The results of the testing show this source to be in compliance with its permitted emission limits of 0.275 pounds of particulate per ton of kiln feed and 133 tons per year of particulate (for both the kiln and the clinker cooler). Since the source's emissions were less than 50% of the permit limit, no further testing is necessary for this point during the five year Title V permit term.

On April 25, 2000, a stack test was performed on the clinker cooler stack. This source was tested to determine compliance with the permit and NSPS particulate emissions limitations. The results of the testing are summarized in the table below.

	Run 1	Run 2	Run 3	Average
Kiln Feed Rate (tons/hr)	112.8	111.5	111.4	111.9
Stack Flow (dscfm)	56700	59600	54600	56900
Oxygen (%)	18.5	18.5	18.5	18.5
Particulate Emissions (gr/dscf)	0.0057	0.0020	0.0096	0.0058
Particulate Emissions (lbs/hr)	2.8	1.0	4.5	2.8
Particulate Emissions (tons/yr)	11.1	4.1	18.2	11.1
Particulate Emissions (lbs/ton)	0.024	0.009	0.40	0.025

Tons per year values in the table above are calculated based on the permit limit of 8064 hours per year of operation. The results of the testing show this source to be in compliance with its permitted emission limits of 0.10 pounds of particulate per ton of kiln feed and 133 tons per year of particulate (for both the kiln and the clinker cooler). Since the source's emissions were less than 50% of the permit limit, no further testing is necessary for this point during the five year Title V permit term.

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#03

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#040

On January 21, 2000 testing was conducted on the Beneficiation Dust System baghouse. The testing was performed to show compliance with the permit and NSPS limits for particulate and opacity. Only one of the two baghouses for the system needed to be tested. The baghouse tested was BH-225-3, which controls the emissions from silo 2844. The results of the testing are summarized in the table below.

	Run 1	Run 2	Run 3	Average
Stack Flow (dscfm)	16100	16000	16000	16100
Oxygen (%)	20.9	20.9	20.9	20.9
Particulate Emissions (gr/dscf)	0.000	0.000	0.000	0.000
Particulate Emissions (lbs/hr)	0.03	0.00	0.01	0.02
Particulate Emissions (tons/yr)	0.13	0.02	0.05	0.07
PM ₁₀ Emissions (gr/dscf)	0.000	0.000	0.000	0.000
PM ₁₀ Emissions (lbs/hr)	0.00	0.00	0.01	0.01
PM ₁₀ Emissions (tons/yr)	0.00	0.02	0.05	0.02
Opacity (%)	0.0	0.0	0.0	0.0

The results of the testing show this source to be in compliance with the permitted emission limits of 5.30 tons per year for total particulate and 2.65 tons per year of PM₁₀. These limits are for both of the baghouses on the system together. The testing was also to be used to verify that PM₁₀ was to verify that PM₁₀ emissions were no more than 50% of total particulate. Although the emissions were so low as to make comparisons difficult, the results do show PM₁₀ to be below 50% of the total particulate emitted. This source does not require an additional test during the five year operating permit term.

Action #041

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